Case Studies

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Task 1: Create and Manage Files

Objective: Practice creating, listing, viewing, and deleting files.

Instructions

1. Create a directory called linux_task1.
2. Enter the directory.
3. Create five empty files:
   - file1.txt
   - file2.txt
   - file3.txt
   - file4.txt
   - file5.txt
4. List all files.
5. Display detailed information about the files.
6. Add some content to file1.txt.
7. Display the contents of file1.txt.
8. Delete file5.txt.
9. List the files again.

Task 2: Copy and Move Files

Objective: Practice copying and moving files and directories.

Instructions

1. Create a directory called linux_task2.
2. Inside it, create:
    - source.txt
    - data.txt
    - notes.txt
3. Add some content to source.txt.
4. Create a directory called backup.
5. Copy source.txt into backup.
6. Copy data.txt to backup and rename it as data_backup.txt.
7. Move notes.txt into backup.
8. Rename source.txt as original.txt.
9. List all files and directories recursively.

Task 3: Working with head and tail

Objective: Practice viewing specific portions of a file.

Instructions

1. Create a directory called linux_task3.
2. Create a file called students.txt.
3. Add at least **15 student names**, one name per line.
4. Display the entire file.
5. Display the first 5 lines.
6. Display the first 10 lines.
7. Display the last 5 lines.
8. Display the last 10 lines.

Task 4: Organize Files

Objective: Organize files into different directories using mv.

Instructions

Start with:

```
organization/
├── file1.txt
├── file2.txt
├── file3.txt
├── image1.jpg
└── image2.jpg
```

Then:

1. Create documents directory.
2. Create images directory.
3. Move all .txt files into documents.
4. Move all .jpg files into images.
5. List the contents of each directory.
6. Rename file1.txt to notes.txt.
7. Rename image1.jpg to photo.jpg.

Task 5: Directory Management

Objective: Practice creating nested directories and removing them.

Instructions: Create the following directory structure:

```
project/
├── source/
│   ├── c/
│   └── python/
├── docs/
└── backup/
```

Then:

1. Create the project directory. 2. Create all subdirectories. 3. Create three files inside source/c. 4. Create two files inside source/python. 5. List the complete directory structure. 6. Remove the empty backup directory. 7. Create a file inside backup. 8. Try to remove backup using rmdir. 9. Observe what happens. 10. Delete the file inside backup. 11. Remove backup using rmdir.

Task 6: File Permissions Using chmod

Objective: Understand Linux file permissions.

Instructions

1. Create a directory called linux_task4.
2. Create three files:
   - public.txt
   - private.txt
   - script.sh
3. Add some content to each file.
4. Display the current permissions using ls -l.
5. Set private.txt permission to 600.
6. Set public.txt permission to 644.
7. Give execute permission to script.sh.
8. Check the permissions again.
9. Remove execute permission from script.sh.

Task 7: File Permission Challenge

Objective: Understand how permissions affect files and directories.

Instructions

1. Create a directory called permission_lab.
2. Create a file called test.txt.
3. Add some content to it.
4. Remove read permission from the file.
5. Try to read the file using cat.
6. Restore read permission.
7. Remove write permission.
8. Try to modify the file.
9. Restore write permission.
10. Remove execute permission.
11. Try to execute the file using ./test.txt.

Task 8: Backup and Restore

Objective: Create backups and restore files using cp.

Instructions

1. Create a directory called linux_task6.
2. Create a file called important.txt.
3. Add at least 10 lines of content.
4. Create a directory called backup.
5. Copy important.txt into backup.
6. Rename the backup as important_backup.txt.
7. Delete the original important.txt.
8. Copy the backup back to the main directory.
9. Rename it to important_restored.txt.
10. Display its contents.

Task 9: Complete Linux File Management Challenge

Objective: Use almost all the commands together.

Create this structure:

```
linux_project/
 ├── source/
│    ├── main.c
│    ├── helper.c
│    └── README.txt
 ├── backup/
 └── docs/
```

Requirements

1. Create linux_project.
2. Create source, backup, and docs.
3. Create the three files inside source.
4. Add 20 lines to README.txt.
5. Display the first 5 lines.
6. Display the last 5 lines.
7. Copy main.c to backup.
8. Copy README.txt to docs.
9. Rename helper.c to functions.c.
10. Change the permissions of README.txt to 644.
11. Delete the copied main.c from backup.
12. Remove the empty backup directory.
13. List the final directory structure.

Task 10: Mini Linux Administration Challenge

Objective: Complete a real-world-style file management exercise.

Imagine you are managing a project called student_portal.

Create:

```
student_portal/
├── src/
├── docs/
├── backup/
├── logs/
└── config/
```

Requirements

1. Create all directories.
2. Create:
   - main.c
   - student.c
   - README.txt
   - config.txt
   - system.log
3. Place the C files inside src.
4. Place README.txt inside docs.
5. Place config.txt inside config.
6. Place system.log inside logs.
7. Add at least 15 lines to system.log.
8. Display the first 5 lines of the log.
9. Display the last 5 lines of the log.
10. Copy README.txt to backup.
11. Rename the backup to README_backup.txt.
12. Change the permission of config.txt to 600.
13. Change the permission of README.txt to 644.
14. Delete README_backup.txt.
15. Remove the empty backup directory.
16. Display the final structure using: ls -R